

UNIT – II: Classes and Objects

Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Constructor Methods for Class, Keyword this. Methods: Introduction, Defining Methods, Overloaded Methods, Attributes Final and Static.

//java program to display the class details

```
class Employee
{
    String name;
    int salary;
    public void display()
    {
        System.out.println("Name is"+name);
        System.out.println("Salary is"+salary);
    }
}
public class Class1
{
    public static void main(String args[])
    {
        Employee raja=new Employee();
        raja.name="Raja Kumar";
        raja.salary=32000;
        raja.display();
    }
}
```

//java program to demonstrate class members

```
class Demo//default class
{
    static int a;
    void increment()
    {
        a++;
        System.out.println("The value of a is"+a);
    }
}
public class ClassMembers
{
    public static void main(String args[])
```

```

    {
        Demo d=new Demo();//declaring object
        d.increment();//a=1
        d.increment();//a=2
        Demo d1=new Demo();
        d1.increment();//a
        d1.increment();
    }
}

```

D:\Openelective2>javac ClassMembers.java

D:\Openelective2>java -cp . ClassMembers

The value of a is1

The value of a is2

The value of a is3

The value of a is4

//java program to implement class and object

```
import java.util.*;
```

```
class Student
```

```

{
    String rollno;
    String name,branch;
    Student()//default constructor method
    {
        rollno="23481A0335";
        name="yasin";
        branch="Mechanical";
    }

    void getStudent()
    {
        System.out.println("The student details are");
        System.out.println("rollno"+rollno+"\nname"+name+"\nbranch"+branch);
    }
}

public class ClassAndObject
{
    public static void main(String args[])
    {
        Student yasin=new Student();
    }
}

```

```

        yasin.getStudent();
    }
}
/*
D:\Openelective2>javac ClassAndObject.java
D:\Openelective2>java -cp . ClassAndObject
The student details are
rollno23481A0335
nameyasin
branchMechanical
*/

```

//java program to implement class and object

```

import java.util.*;
class Student
{
    String rollno;
    String name,branch;
    Student(String r, String n,String b)//parameter constructor method
    {
        rollno=r;
        name=n;
        branch=b;
    }

    void getStudent()
    {
        System.out.println("The student details are");
        System.out.println("rollno"+rollno+"\nname"+name+"\nbranch"+branch);
    }
}
public class ParameterizedConstructor
{
    public static void main(String args[])
    {
        Student yasin=new Student("23481A0335","YASIN","mech");

        yasin.getStudent();
    }
}

```

```

}
/*
D:\Openelective2>javac ParameterizedConstructor.java
D:\Openelective2>java -cp . ParameterizedConstructor
The student details are
rollno23481A0335
nameYASIN
branchmech
*/

```

//java program to implement this keyword.

```

class Box
{
    double width;
    double depth;
    double height;
    Box(double width,double depth,double height)//parameter constructor
    {
        this.width=width;
        this.depth=depth;
        this.height=height;
    }
    double volume()
    {
        return width*depth*height;
    }
    void showBox()
    {
        System.out.println("The Box Details is");
        System.out.println("The width is"+width);
        System.out.println("The depth is"+depth);
        System.out.println("The height is"+height);
        System.out.println("The volume of box is"+this.volume());// this method
call
    }
}
public class ThisDemo
{
    public static void main(String args[])
    {

```

```
Box b=new Box(1,2,3);
b.showBox();
```

```
    }
}
```

```
/*
```

```
D:\Openelective2>javac ThisDemo.java
```

```
D:\Openelective2>java -cp . ThisDemo
```

The Box Details is

The width is1.0

The depth is2.0

The height is3.0

The volume of box is6.0*/

//java program to demonstrate method overloading

```
class Overload
```

```
{
```

```
    public void add(int a,int b)
```

```
    {
```

```
        System.out.println("The add of two int is" +(a+b));
```

```
    }
```

```
    public void add(float a,float b)
```

```
    {
```

```
        System.out.println("The add of two float is" +(a+b));
```

```
    }
```

```
    public void add(int a,float b)
```

```
    {
```

```
        System.out.println("The add of two int is" +(a+b));
```

```
    }
```

```
    public void add(float a,double b)
```

```
    {
```

```
        System.out.println("The add of two int is" +(a+b));
```

```
    }
```

```
    public void add(int a,float b,double d)
```

```
    {
```

```
        System.out.println("The add of two int is" +(a+b+d));
```

```
    }
```

```
}
```

```
public class MethodOverloading
```

```
{
```

```

    public static void main(String args[])
    {
        Overload o=new Overload();
        o.add(1,2);
        o.add(1,2.5f);
        o.add(2.5f,5.6);
        o.add(1,2.3f,4.5);
    }
}
/*
D:\Openelective2>javac MethodOverloading.java
D:\Openelective2>java -cp . MethodOverloading
The add of two int is3
The add of two int is3.5
The add of two int is8.1
The add of two int is7.799999952316284*/

```

//java program to access private members of a class

```

class Demo
{
    private int a;
    private int b;
    Demo(int aval,int bval)
    {
        a=aval;//1
        b=bval;//2
    }
    public void showDemo()
    {
        System.out.println("The value of a is "+a);
        System.out.println("The value of b is "+b);
    }
}

public class AccessPrivateMembers
{
    public static void main(String args[])
    {
        Demo d=new Demo(1,2);
        d.showDemo();
    }
}

```

```
}
/*
D:\Openelective2>javac AccessPrivateMembers.java
D:\Openelective2>java -cp . AccessPrivateMembers
The value of a is 1
The value of b is 2*/
```

//java program to demonstrate methods

```
class Farm
{
    int length;
    int width;
    public void area()
    {
        int a;
        a= length*width;
        System.out.println("The area of the farm is"+a);
    }
}

public class MethodDemo
{
    public static void main(String args[])
    {
        Farm f=new Farm();
        f.length=150;
        f.width=200;
        f.area();
    }
}

/*D:\Openelective2>java -cp . MethodDemo
The area of the farm is 30000*/
```